RTSfM: Real-Time Structure From Motion for Mosaicing and DSM Mapping of Sequential Aerial Images With Low Overlap

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***Abstract*— Inspired by simultaneous localization and mapping (SLAM) style workflow, this article presented an online sequential structure from motion (SfM) solution for high-frequency video and large baseline high-resolution aerial images with high efficiency and novel precision. First, as traditional SLAM systems are not good in processing low overlap images, based on our novel hierarchical feature matching paradigm with multihomography and BoW, we proposed a robust tracking method where the relative pose and its scale are estimated separately followed by a joint optimization by considering both perspective-n-point (PnP) and epipolar constraints. Second, to further optimize the camera poses for the sparse map and dense pointcloud reconstruction, we provided a graph-based optimization with reprojection and GPS constraints, which make the camera trajectory and map georeferenced. We also incrementally generated the dense point cloud in real time from keyframes after local mapping optimization. Finally, we use a publicly available aerial image dataset with sequences of different environments, to evaluate the effectiveness of the proposed method, meanwhile, the robust performance of our solution is demonstrated with applications of high-quality aerial images mosaic and digital surface model (DSM) reconstruction in real time. Compared with the state-of-the-art SLAM and traditional SfM methods, the presented system can output large-scale high-quality ortho-mosaic and DSM in real time with the low computational cost.**

***Index Terms*— 3D reconstruction, aerial photogrammetry, simultaneous localization and mapping (SLAM), terrain mapping, unmanned aerial vehicles (UAVs).**

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I. Introduction

In RECENT years, unmanned aerial vehicles (UAVs) have become increasingly popular because they can accomplish aeronautic tasks at a low price. One of the most popular tasks is the use of UAVs in collecting aerial images for low-cost photogrammetric surveys, which are often the basis of topographic maps, land-use planning, and environmental studies. Real-time high-quality aerial images mosaic and digital surface model (DSM) reconstruction [1] can further help missions such as fire monitoring [2], [3] and urban surveillance [4] where both efficiency and accuracy are required.

Structure from Motion (SfM), which is designed to explore the most information and reconstruct a metric model from unordered multiview images, is a mature choice for photogrammetric surveys. Early self-calibrating reconstruction systems are built to compute 3-D structures from the unordered 2-D images of urban scenes [5]–[7]. Subsequently, various incremental [8]–[10], global [11], [12], and hierarchical SfM strategies [13] have been proposed for different considerations. Inspired by these works, increasingly large-scale reconstruction systems, such as open-source projects (e.g., OpenMVG [10] and openDroneMap) and commercial software (e.g., Pix4DMapper [14] and Photoscan [15]), have been developed. These systems share almost the same pipeline: feature extraction and matching, image alignment, and sparse point cloud generation with bundle algorithm [16], dense point-cloud, and additional applications. Although these systems have advanced state-of-the-art robustness, accuracy, and completeness, they often need all N images prepared to apply exhaustive feature extraction and matching with $O(N^2)$ complexity; They must also spend hours or even days to produce a reconstruction of the observed camera trajectory and scene structure, especially for high resolution and low overlap aerial images taken sequentially. Therefore, they are not suitable for real time and incremental usage.

Real-time visual simultaneous localization and mapping (SLAM) is a popular research topic in the robotics and computer vision community. Unlike traditional SfMs that are applied for feature extraction, matching, and step-by-step image alignment, the processing flow of SLAM approaches are well designed to achieve real-time performance. In recent years, several keypoint-based [17]–[19], direct [20]–[22], and semidirect monocular visual SLAM systems [23], [24] have been proposed and developed rapidly. Given that monocular

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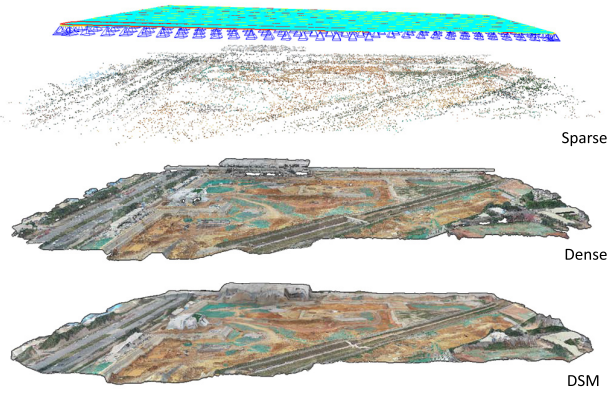


Fig. 1. RTSfM presents a real-time system to reconstruct sparse reconstruction, dense pointcloud, and DSM (ortho-mosaic and elevation image) incrementally at the same time. Both high-overlap video and low-overlap images are supported, and GPS information is fused for generating georeferenced results.

SLAM systems lack scale information and cannot handle pure rotation situation, other multisensor SLAM systems including RGBD [25]–[27], stereo [24], [28], [29], and inertial aided methods [30]–[32] are being investigated for high robustness and precision. These SLAM systems show high efficiency in processing high-frequency videos. However, they cannot handle low overlap (large baseline) high-resolution images yet, and do not meet the demands of high-quality aerial images mosaic and DSM reconstruction in real time. Recently, supervised [33]–[35] and unsupervised [36], [37] deep learning-based visual odometer (VO) present novel results compared with traditional geometry-based methods. However, at present, the former is restricted to partial situations with low generalization and poor stability.

In sum, both SfM and SLAM have advantages and disadvantages. SfM is slow for real time and incremental usage, but it is able to provide accurate 3-D information from a series of unordered images without any additional information. SLAM can hardly work unless the initial value is close to the real value, especially for low overlap images, but it is fast. The approach proposed in this article combines SfM and SLAM. It uses SfM to achieve accurate results, and uses the serialized strategy of SLAM to accelerate SfM. In this article, our goal is to develop an online incremental SfM method for high-frequency video and large baseline high-resolution sequential aerial images. Given that SfM methods are fundamentally offline in nature and processing low-overlap images using direct SLAM methods is still very difficult, we introduced the popular feature-based SLAM workflow into SfM technology, where tracking and mapping are separated in two threads and we do not need to match every neighbor before localization. For small baseline video, map-points are matched by projection with a motion model pose forecast followed by a nonlinear PnP optimization which is commonly used by existing SLAM systems and shows efficiency and accuracy. For large baseline images, a global matching between the current frame and the last keyframe is accelerated using the multihomography and pretrained bag-of-words tree [38]. We estimated the pose using RANSAC-based PnP. However, this method sometimes fails or outputs a poor estimation because insufficient map-points are found, and most of the projection lies on the same area of the image. To solve this problem, this article proposes a two-step method for the robust and precise tracking of low-overlap images. Because the proposed method has merits of real-time and fully automatic properties, it has a good prospect for applications such as high-quality aerial images mosaic and DSM reconstruction in real time.

In brief, the main contributions are summed up as follows:

1) *An Online Sequential Structure From Motion (RTSfM) Framework With High Efficiency and Novel Precision, Where Georeferenced Sparse Reconstruction, Dense Pointcloud, and DSM (Ortho-Mosaic and Elevation Image) Are Generated Incrementally at the Same Time:* While traditional SfM-based reconstruction system are naturally designed for offline unordered images, this article presents a system providing real-time performance to handle live high-resolution images and outputs high-quality dense pointcloud along with DSM. This system has great potential by integrating to onboard computers and providing live reconstruction during the flight.
2) *A Robust Georeferenced Tracking System for Both High-Frequency Video and Large Baseline High-Resolution Aerial Images:* High-frequency video streams or high-resolution images are two main resources for aerial photogrammetry survey systems. While most traditional SLAM systems demand a low resolution with high-frequency, we exploit a novel georeferenced tracking system that is able to handle low overlap high-resolution images while retaining the ability to process high-frequency video streams. A multihomography and BoW accelerated matching system is used to provide robust global matches efficiently. Instead of only using PnP constraints, we estimate the pose of large-baseline images with a joint optimization by considering both PnP and epipolar constraints, and GPS information are further fused for obtaining georeferenced localization and correct scale.
3) *An Efficient Incremental Densification and Tiled DSM Generation Pipeline:* While traditional multiview stereo, meshing, and texturing approaches are too computationally expensive to satisfy live reconstruction demands. This article presents an incremental dense pointcloud and textured terrain generation system, which is able to handle large-scale high-resolution images lively with the low computational cost. The outputs are all georeferenced and the experiments show that both our dense pointcloud and DSM are comparable to state-of-the-art offline softwares.

Notably, this article extends our early conference work [1] through the following aspects. 1) Beyond performing high-quality aerial images mosaic [1], we provide another sequential application implementation of dense pointcloud and DSM reconstruction in real time, to evaluate the effectiveness of the proposed method. The presented system can output large-scale high-quality pointcloud, ortho-mosaic, and DSM in

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real time with the low computational cost. 2) By the upgraded algorithm, we extend the input resource to large-baseline high-resolution images. An efficient BoW-based multihomography matching algorithm is used to provide reliable matches, and a robust tracking pipeline by fusing PnP, epipolar, and GPS constraints is proposed to obtain precise incremental sparse reconstruction. 3) We present a novel live densify module with efficient filtering to generate high-quality dense pointcloud while keeping the ability to refine the pose along with local bundle adjustment (BA). A large-scale DSM fusion system is demonstrated to generate high-resolution ortho-mosaic and elevation model, and an efficient georeferenced rendering engine is developed to display the live terrain by using the tiled LoD technique. 4) Extensive experiments and analyses on larger datasets are conducted to verify the effectiveness of the system.

We compare our method with the current state-of-the-art SLAMbased system (ORBSLAM2 [29]), SfM-based systems (COLMAP [8], OpenMVG [10]), and Mosaicing systems (Map2DFusion [1], TerrainFusion [39], and OpenREALM [40]). The evaluation result demonstrates that our system achieves novel efficiency and quality on video and high-resolution image sequence processing.

Fig. 1 demonstrates that our method on handling a dataset is composed of 300 images, our real-time system can reconstruct sparse reconstruction, dense point cloud, and DSM (ortho-mosaic and elevation image) incrementally at the same time. Both high-overlap video and low-overlap images are supported and GPS can be fused to generate georeferenced results. A demo video of our system can be found at: https://youtu.be/7QB-O4GJmgI.

II. Related Work

Many works have been conducted to do photogrammetric surveys in recent years, the proposed approach takes advantages of both SfM and SLAM technologies, which develop rapidly and some representative works are summarized below.

A. SfM-Based Methods

SfM is a central problem in computer vision community for estimating 3-D structure from image collection [5]–[7] on different positions. It is commonly used as the base technology for photogrammetric surveys [10], [14], [15], [41], [42]. So far, various SfM strategies including incremental [8], [9], global [11], [12], and hierarchical approaches [43], have been proposed.

Incremental SfM [8], [9] is a prevalent strategy for 3-D reconstruction from unordered image collections. It is a sequential processing pipeline [44] that adds on one image at a time to grow the reconstruction gradually. Incremental SfM usually begins from two-view reconstruction, registers cameras iteratively, and performs BA [16] for each registration to refine the camera poses and scene structure. In [13], they proposed an efficient nonconsecutive feature tracking (ENFT) framework to match interrupted tracks distributed in different subsequences or even in different videos. A novel scheme [45] uses meta-data information such as GPS, keypoint filtering, and multiple local BA refinement instead of global optimization to speed up the incremental SfM process. Although incremental SfM is robust in reconstructing complete 3-D sparse point scenes, it may suffer from drift, inefficiency and poor scalability when used for large-scale scene reconstruction, because it requires the repeated operations of exhaustive feature matching and expensive BA.

Global SfM [11], [12] is different from incremental SfM because it considers the entire view graph at the same time instead of adding more and more images to the reconstruction incrementally. Global SfM methods have been proven to be very fast with comparable or better accuracy to incremental SfM approaches [46], [47], and they are much more readily parallelized. After obtaining camera poses, global SfM performs triangulation and BA to obtain a valid 3-D reconstruction that consists of cameras and 3-D points. It is worth noting that a significant defect of incremental SfM is the accumulative error, the reconstruction of the final results will drift in the loop closure scene. Global SfM can overcome this problem by considering the spatial relationship between the adjacent cameras. However, global SfM relies on correct matching results excessively, and reconstruction precision and completeness are worse than incremental SfM.

Hierarchical SfM [48] generally organize images into a tree with agglomerative clustering, using a measure of overlap as the distance. The reconstruction follows this tree from the leaves to the root. As a result, the problem is broken into smaller instances, which are then solved and combined separately. Hierarchical SfM has a lower computational complexity compared to incremental SfM and global SfM. It is independent of the initial pair of views, and copes better with drift problems. Building on [48], improved hierarchical SfM [43] introduced a clustering strategy derived from the simple linkage that makes the dendrogram more balanced.

Moreover, the reconstruction of manifold surface [41] can utilize 3-D Delaunay triangulation to reconstruct surface from sparse 3-D points calculated by using the aforementioned SfM methods. However, [41] aims to reconstruct a closed surface, which makes it unsuitable for DSM reconstruction. Poisson surface reconstruction [42] is a famous 3-D reconstruction approach that expresses surface reconstruction as the solution to a Poisson equation. Although Poisson reconstruction approach shows a mastery of technical skill and detail, it cannot reconstruct a surface in real time.

B. SLAM-Based Methods

SLAM is regarded as the core to realize navigation and environment sensing for fully autonomous robotics. Parallel Tracking and Mapping (PTAM) [18] is a famous monocular SLAM implementation that shows high efficiency but is limited to small workspaces. Given that most methods detect keypoint features for matching, Newcombe *et al.* [20] propose a direct approach named DTAM, wherein camera motion was computed by minimizing a global spatially regularized energy function on GPU. To reduce the intensively computational demand of DTAM, Forster *et al.* [23] proposed a semidirect monocular visual odometer (SVO) which only processed strong gradient pixels and resulted in high frame rates. Although all these methods only track camera motion locally

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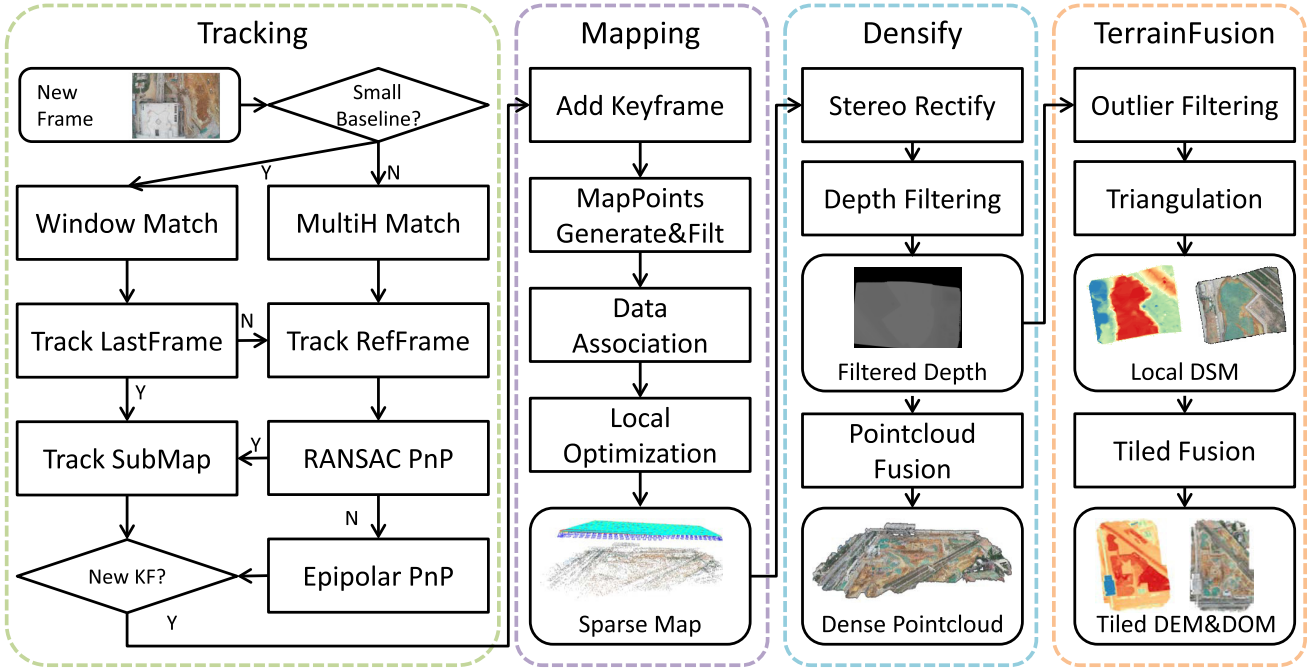


Fig. 2. Framework of RTSfM with four working modules. A classic SLAM pipeline is used to separate the tracking and mapping works for sparse reconstruction generation. The densify module explores stereo image pairs for depth estimation and fuses them to a global keyframe referenced pointcloud. The TerrainFusion module estimates local elevation and orthophoto patches, and fuses them to a tile cached DSM manager. Our system can output sparse reconstruction, dense pointcloud, and texturing DSM in real time.

and did not have loop-closures, Engel *et al.* [21] showed a well-known large-scale direct monocular SLAM (LSD-SLAM). The approach utilizes FabMap [49] for loop detection and a similarity transform pose graph for optimization to build consistent, large-scale maps of the environment. It is extended to omnidirectional and stereo cameras [28], [50].

Direct methods rely on small baselines and seem to lack robustness when the frame-rate is not high enough. Mur-Artal *et al.* [19] proposed a novel ORB feature-based monocular SLAM algorithm (ORB-SLAM), that could handle large baselines robustly with high precision. It is also further explored for semidense reconstruction and obtains better results than direct methods [51]. To improve tracking accuracy, Bu *et al.* [27] proposed a semidirect tracking and mapping (SDTAM) method for RGB-D device, which inherits the advantages of direct and feature-based methods. Based on the visual SLAM algorithms, several works proposed to perform live aerial survey where incremental ortho-mosaic [52] and elevation [39], [40] can be produced during the flight.

Recently, end-to-end approaches based on convolutional neural networks have become popular. Supervised [33]–[35] and unsupervised [36], [37] deep learning-based VO present novel ideas compared with traditional SLAM-based methods. However, optimizing the predicted poses further for consistencies of multiple keyframes is still not easy. Because these methods learn to regress the camera pose from an input image [53] directly, which do not achieve the same level of pose accuracy as 3-D structure-based methods.

III. Our Approach

A. Framework

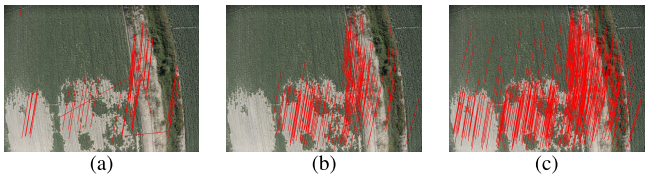
Inspired by the classic SLAM pipeline [54], a novel real-time structure from motion (RTSfM) is used for frame tracking, sparse point cloud map estimation, dense point cloud reconstruction, and high-quality DSM generation, as demonstrated in Fig. 2.

Since direct SLAM methods require higher overlap between frames, a feature-based method similar with [29], [55] is adopted. Rather than using binary descriptors such as BRIEF [56] and ORB [57], which are not suitable for high resolution and low overlap image matching, a GPU accelerated scale-invariant feature transform (SIFT) implementation is used in this framework. The map is initialized with relative pose decomposition from fundamental or homography matrix followed by a two-frame georeferencing using GPS information joint optimization. To handle both video and large-baseline images, different strategies are used to ensure robust tracking and map point triangulation. A timestamp check is used to decide whether small or large baseline tracking should be launched. Section III-B will describe the details for small baseline tracking. Sections III-C and III-D will introduce the proposed global frame matching and large baseline tracking approach. Keyframes are selected for new map-points generation and later local optimization with GPS which prevents drift after a long trajectory. The details of mapping will be described in Section III-E, and the latest keyframe with pose estimated and sparse depth information is published for later online DSM fusion work, which is described in Section III-F.

B. Small Baseline Video Tracking Pipeline

For every incoming frame, if the time difference against the preview frame is not greater than one second, then a small baseline strategy is used. The motion prior is used to provide a first guess of current camera pose, with an assumption that the UAV velocity do not changes too quickly.

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(a) (b) (c)

Fig. 3. Sample matching flow result comparison. (a) Matching result of ORBSLAM2, where BoW is used to accelerate global matching. (b) Result with BoW feature matching and filtered with RANSAC. (c) Final result of our algorithm where better matches are found. (a) ORB. (b) RANSAC. (c) RANSAC+MultiH.

By projecting the last tracked map-points to the current frame, the corresponding keypoints should lie in a windowed area. Thus, window searching between the current frame and the last tracked map-points is applied to collect several perspective-n-point (PnP) matches.

For small baseline sequences, the camera pose $\mathbf{T}_i = (\mathbf{R}_i, \mathbf{t}_i)$ can be estimated by using the PnP constraints only. Given a set of matches between 3-D map-points $\mathbf{P}$ and current keypoints $\mathbf{u}$, our target is to minimize the weighted squared reprojection error in the Lie algebra manifold space

$$\mathbf{T}_i^* = \arg\min \sum_{\mathbf{P}_j \in \mathbf{P}} w_j \mathbf{e}_j^T \Sigma_j^{-1} \mathbf{e}_j. \tag{1}$$

Here, the information matrix Σ_j^{-1} is determined by the size of keypoint, and weight w_j is given by a Huber robust weighted function to reduce the influence of large residuals. We denote $\mathbf{u} = \pi(\mathbf{P})$ as the camera projection function, the error $\mathbf{e}_j$ is defined as the reprojection difference between keypoint position measurements $\mathbf{u}_j = (x_j, y_j)^T$ and predicted $\mathbf{u'}_j$ of a map-point $\mathbf{P}_j$

$$\mathbf{e}_j = \mathbf{u}_j - \mathbf{u'}_j = \begin{bmatrix} x_j \\ y_j \end{bmatrix} - \pi(\mathbf{T}_i \cdot \mathbf{P}_j). \tag{2}$$

After the camera pose is optimized by the above cost function, all keyframes observing these map-points are collected to construct the local submap, and more matches are found by projecting all local map-points with the coarse pose. The final camera pose is estimated by optimizing 1 again with all local map-points observations. It is noted that, the pose optimization in 1 is performed with iteratively reweighted least squares (IRLS) and we use the open-source Ceres Solver library[1] to solve this nonlinear graph optimization problem in our implementation.

C. Multihomography and BoW Accelerated Matching

When the overlap between two neighbor frames is not enough to utilize a window match searching, a global matching strategy is required. For n descriptors in each frame, the brute force matching has $O(n^2)$ complexity and that of FLANN [58] is $O(n \log n)$. ORB-SLAM [19] uses a pretrained BoW vocabulary to split descriptors into small spaces to further decrease the computational complexity, and every space can perform matching separately. The computation is dramatically decreased to even near $O(n)$ complexity when a proper level is selected to split the descriptor spaces.

[1]http://www.ceres-solver.org/

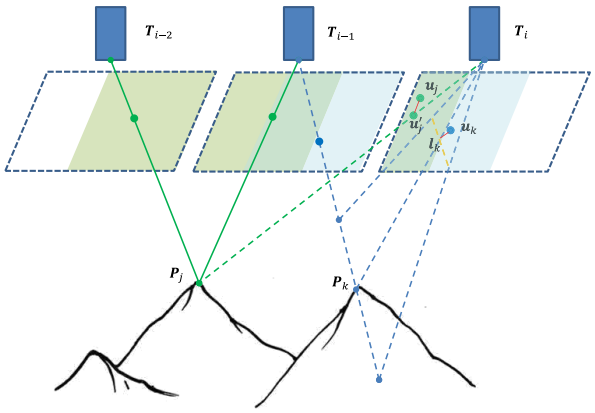


Fig. 4. Geometry description of PnP and epipolar errors. The green area corresponds to the overlap between frame T_{i-2} and T_{i-1}, whereas these two frames triangulated map-point P_j. The blue area refers to the overlap between frame T_{i-1} and T_i, where P_j is projected to u'_j, and the reprojection error provides the PnP constraint. 3-D point P_k is unknown and corresponds to epipolar line l_k when estimating pose T_i. The epipolar constraints are obtained by minimizing the distance between line l_k and keypoint measurement u_k.

In this article, the splitting strategy is adopted to obtain an initial matching, and the accelerated version of DBoW implemented by GSLAM [54] is used. A fundamental matrix is estimated on the basis of epipolar geometry with the random sample consensus (RANSAC) [59] procedure for geometric outlier removal because outliers still exist in the initial matches.

The above pipeline should have matched two images successfully, however, a good match not always lies in the same branch split by the BoW tree, that is, some good matches are ignored. Sometimes, a good match does not hold the minimum descriptor distance. To further improve the matching results, RTSfM presents a multihomography way to rematch those potential inliers.

Multiple homography matrices are estimated iteratively using the previous inlier matches. Then, for unmatched features, several target positions are calculated using these homography estimations. The distance between target positions and the epipolar line is computed to select the best target position near the epipolar line. When the distance is below the threshold, window searching is performed to find its potential matches. The search radius is the distance between the epipolar line and the best target position, whereas a rematch is obtained when the descriptor distance is below the median matching distances.

As demonstrated in Fig. 3, providing robust matching results for such low-overlap high-resolution images is difficult for the ORB descriptor. Our rematching strategy successfully obtained several good matches with very low extra computational cost, and the whole matching procedure only takes less than 3 ms for each image pair. The match information will not only be used by the tracking module, but will also be reserved in the connection data structure between those two frames.

D. Pose Estimation With PnP and Epipolar Constraints

As for SLAM tracking, the PnP algorithm is popularly used for estimating the real-time pose of the camera. However, when the overlap is low, the PnP constraints may not estimate a

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sufficienty good pose because all keypoints lie in the small part of the area, and matches without map point observations are not used. As illustrated in Fig. 4, when the forward overlap is 60%, only less than 20% matches provide PnP constraints. The poor estimation of tracking will result in triangulation failure, because the tracking and mapping steps of SLAM are closely related. When the overlap is enough, a low failure rate is acceptable, and localization can still work with existing map-points. Unfortunately, the tracking will be lost immediately once the localization fails, and the overlap is low and no map-points are triangulated before the later frame coming. Thus, the robustness of tracking and map-points triangulation must be ensured, especially for low-overlap sequences.

To guarantee the robustness of large-baseline tracking, the final pose should be precise enough for triangulation and satisfy epipolar consistency. This work proposes to estimate the camera pose by considering both PnP and epipolar constraints, where matches without corresponding map point also bring constraints and help improve localization robustness. A reference keyframe is first selected, and when the tracking is good, the last keyframe is treated as the reference frame, otherwise the reference frame is selected from neighbor images according to GPS or the pretrained BoW tree. A global matching between current and reference frame is performed, and matches both with or without map-point observation are obtained. The keypoints with observation are regarded as PnP constraints and the traditional projection error is used. For matches without valid observations, they also provide the epipolar constraints and we try to minimize the distance between keypoint against their corresponding epipolar line. The final pose $\mathbf{T}_i^*$ is optimized by minimizing the reprojection error $\mathbf{e}_j$ and epipolar distance error d_k

$$\mathbf{T}_i^* = \arg\min \sum_{\mathbf{P}_j \in \mathbf{P}} w_j \mathbf{e}_j^T \Sigma_j^{-1} \mathbf{e}_j + \sum_k w_k \sigma d_k^2 \tag{3}$$

$$\mathbf{l}_k = \begin{bmatrix} a_k \\ b_k \\ c_k \end{bmatrix} = \mathbf{K}^{-T} \mathbf{R}_{i-1,i}^T (\mathbf{t}_{i-1,i} \otimes \pi^{-1}(\mathbf{u}_k)) \tag{4}$$

$$d_k = \frac{a_k \cdot x_k + b_k \cdot y_k + c_k}{\sqrt{a_k^2 + b_k^2}}. \tag{5}$$

We define $\otimes$ as the cross operation of two vectors. Here w_j and w_k are the Huber robust weights, and the epipolar line $\mathbf{l}_k$ can be derived from relative transform $\mathbf{T}_{i-1,i} = (\mathbf{R}_{i-1,i}, \mathbf{t}_{i-1,i})$ between the current and the reference frame. Since the pose of the reference frame T_{i-1} has been known, the relative pose can be represented by $\mathbf{T}_{i-1,i} = \mathbf{T}_{i-1}^{-1} \cdot \mathbf{T}_i$. As illustrated in Fig. 4, here d_k is the signed distance error between epipolar line $\mathbf{l}_k$ and keypoint $\mathbf{u}_k$.

When the corresponding GPS position and gimbal orientation $\mathbf{g}_i$ are available, an absolute 6 degree of freedom (DOF) pose prior constraint is further used to help tracking, which is defined as

$$\boldsymbol{\xi}_i(\mathbf{T}_i, \mathbf{T}_i^{gps}) = \log(\mathbf{T}_i^{-1} \cdot \mathbf{T}_i^{gps}). \tag{6}$$

Here $\boldsymbol{\xi}_i \in R^6$ is the Lie algebra representation of transform difference. This factor can also work when gimbal rotations are not available and only GPS is provided by adjusting the information matrix. This GPS constraint helps especially when the overlap is very low and PnP constraints are not enough. The epipolar observations can fix the relative pose and GPS decides the scale.

E. Reprojection and GPS Constraints for Map Optimization

Although the tracking module estimates the pose of every image, the mapping module runs in a separate thread and our implementation uses the messenger in GSLAM [54] framework to decouple it from other modules. After the tracking module publishes the keyframes, the mapping module receives them and optimizes the reconstruction for later tracking. First, new map-points should be created and added to the reconstruction as soon as possible, so that enough PnP correspondences are able to be matched for tracking. Matched information reserved in connections is used to triangulate new map-points. The correspondences of other keypoints that are not observed in tracking in the neighbor keyframes along the epipolar line is matched with a bag-of-words (BoW) accelerate searching. After new map-points are created, a data association step is performed to add more observations, combine redundancy map-points, and remove poor map-points.

To further optimize the camera poses and triangulated map-points with the obtained observations, local optimization is performed by considering reprojection and GPS constraints. The GPS constraints can prevent drift and maintain global pose precision effectively. However, sensor fusion requires good calibration, and time should be synchronized. When GPS is not available, a general local BA, such as ORBSLAM2, is performed

$$\{\mathbf{T}_i^*, \mathbf{P}^*\} = \arg\min \sum_{\mathbf{T}_i} \sum_{\mathbf{P}_j \in \mathbf{P}} w_j \mathbf{e}_j^T \Sigma_j^{-1} \mathbf{e}_j + \sum_{\mathbf{T}_i} \boldsymbol{\xi}_i^T \Sigma_i^{-1} \boldsymbol{\xi}_i. \tag{7}$$

F. Incremental Dense Point Cloud Reconstruction

A densification procedure is required to obtain points because the sparse map-points generated from keypoints are not enough to represent the reconstruction details. For SfM pipelines, multiple view stereo (MVS) is popularly used in commercial software (e.g., Pix4DMapper) or open-source systems (e.g., VisualSfM, OpenSfM, and COLMAP [8]). However, MVS is very computationally expensive and often taken after the reconstruction is finished, which is difficult to perform in real time. To reduce computational expense, the depth filter is proposed in direct or semidirect SLAM systems. However, these methods are designed for small movement videos and need very high overlap between frames, which do not meet our demands.

In most real-time aerial mapping circumstances, the camera is keeping down-look and flying forward, which means stereo image pairs can be rectified once the camera poses are known. We present an efficient and robust way to generate and fuse the depth map to a clean point cloud by constructing virtual stereo image pairs. Keyframes are published in the mapping module after local optimization, and we try to select the best neighbor image for the latest keyframe to form proper stereo images. The stereo pair should contain less relative rotation and enough translation according to the flight height. For the

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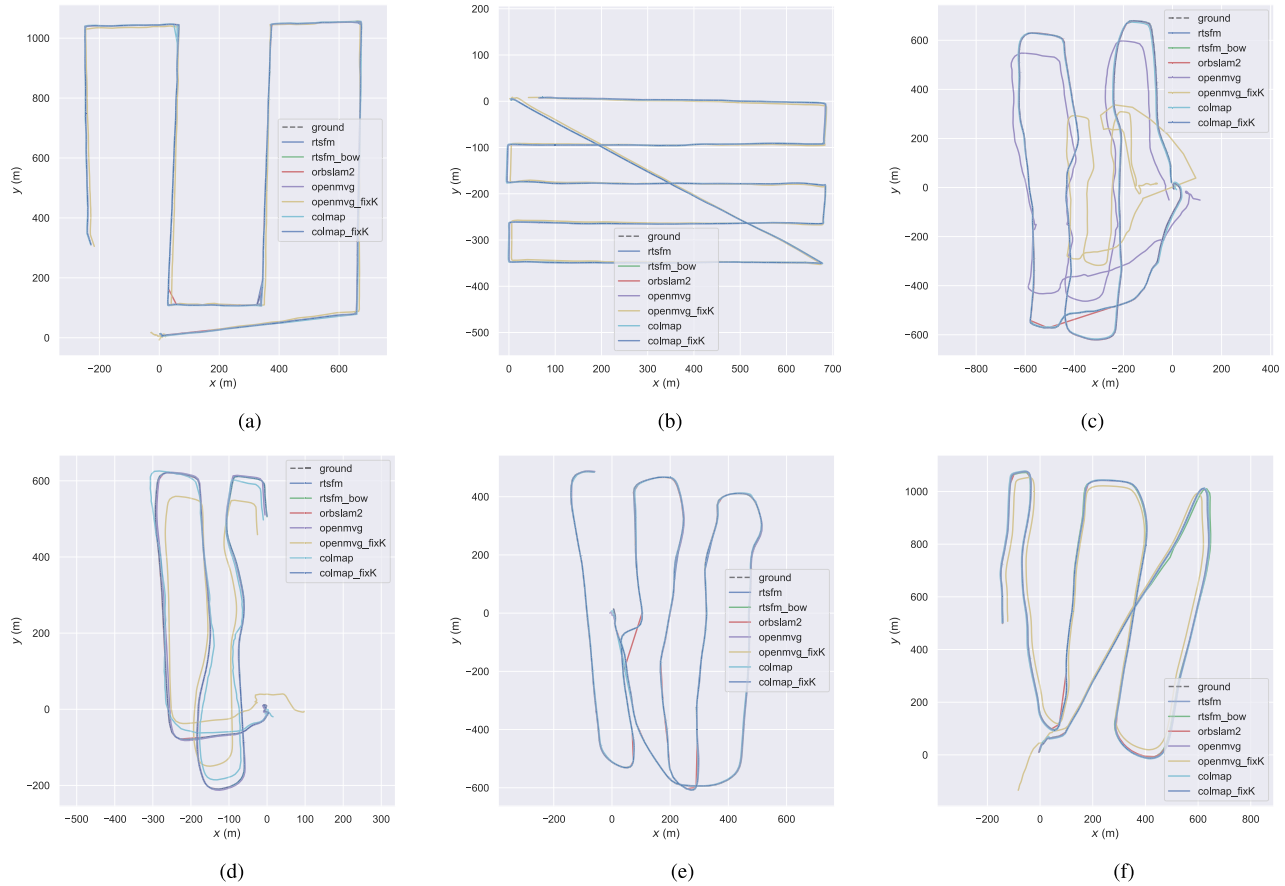


Fig. 5. Trajectories are generated with different systems from monocular images and compared with the ground GPS trajectory. Most of them are well aligned and our system achieves approximately 1/1000 absolute position error without the help of GPS information. The global reconstruction pipeline of OpenMVG performs poor in sequence *phantom3-freeway*, *phantom3-highflower*, and *phantom3-npu*, while COLMAP shows significant drift in sequence *phantom3-highflower* when calibration parameters are not fixed. (a) gopro-npu. (b) gopro-saplings. (c) phantom3-freeway. (d) phantom3-highflower. (e) phantom3-huangqi. (f) phantom3-npu.

selected stereo pair, a rectification step is taken to undistort the images and remap the pixels. Densification after correction is more efficient because all epipolar lines are horizontal and easier to match. Depth directly corresponds to the disparity image. Depth estimation for stereo cameras is a very active research and we use a method called ELAS [60] (semi-global matching (SGM) or SGBM [61] can also be used), which is suitable for fast stereo matching of high-resolution images. This method builds a prior on the disparity by forming triangulations on a set of support points which can be matched robustly, thereby reducing the matching ambiguities of the remaining points. This allows the efficient exploitation of the disparity search space, thereby yielding accurate and dense reconstructions without global optimization.

For the depth from disparity, there are still outliers that should be filtered. Consistency check and refinement are performed by reprojecting the current depth map to its neighbors. Keyframes with cleaned depth information will be published for the later DSM reconstruction, and redundant points are pruned and fused to the global pointcloud manager for visualization and processing. The corresponding point subset should be rendered according to the pose of its reference keyframe because the keyframe poses are not static and will be adjusted after local optimization.

G. Online Applications of Mosaicing and DSM Fusion

A digital orthophoto generation from aerial images is important in various applications and is the core functionality of commercial software, such as Pix4DMapper and PhotoScan, which need all images to be prepared and require a long offline processing time. Map2DFusion [1] proposed an online mosaicing method based on monocular SLAM and showed high quality. However, it was based on Homography projection, which was only suitable for planar scenarios and could not output measurable true orthomosaic. To resolve this problem, Wang *et al.* [39] proposed TerrainFusion which could generate online DSM incrementally. However, TerrainFusion only deals with the sparse point cloud. Showing the building details is difficult, and the resolution of the result is low. Here, we extend Map2DFusion and TerrainFusion with dense depth information and improve the local DSM generation with a more efficient triangle rendering algorithm from OpenGL. Multithread processing is supported to further accelerate fusion performance and a hashed least recently used (LRU) cache is applied for large-scale ability by caching tiles to the disk. We also optimized the online rendering of a large-scale DSM by levels of detail (LOD) technology and only visible tiles are rendered according to the projection matrix.

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TABLE I

COMPARISON OF ATE OF THE PROPOSED METHOD WITH ORB-SLAM2, COLMAP AND OPENMVG ON THE NPU DRONEMAP DATASET. THE MEASURE METRIC IS ROOT-MEAN-SQUARE-ERROR (RMSE) AND THE UNIT IS THE METER (M). IT SHOULD BE NOTED THAT FOR TRADITIONAL SFM METHODS, THE INTRINSIC REFINEMENT DOES NOT BRING BETTER PRECISION SINCE THE FOCAL CAN HARDLY BE FIXED BY AUTOMATIC CALIBRATION OF BA WHEN ONLY USING THE DOWN-LOOK IMAGES

Sequence Name	RTSfM(full)	RTSfM(bow)	ORB-SLAM2	COLMAP(fixK)	COLMAP	OpenMVG(fixK)	OpenMVG
gopro-monticules	0.968	1.239	1.354	0.889	7.140	**0.767**	1.493
gopro-npu	**0.491**	0.587	0.599	0.522	9.086	3.662	5.629
gopro-saplings	0.970	1.632	1.057	**0.435**	6.615	76.391	2.182
phantom3-factory	2.88	3.581	3.232	**2.329**	9.674	38.507	3.517
phantom3-freeway	**1.189**	1.290	1.924	1.253	17.279	181.469	9.291
phantom3-highflower	0.422	0.616	0.545	**0.252**	31.480	1.290	2.042
phantom3-huangqi	0.523	0.683	0.754	**0.432**	9.729	0.759	4.756
phantom3-ieu	0.794	0.943	0.927	**0.510**	11.928	16.970	12.990
phantom3-npu	1.271	4.257	2.301	**1.131**	9.573	67.681	5.620
phantom3-village	0.647	0.778	0.764	**0.534**	8.956	1.543	2.930

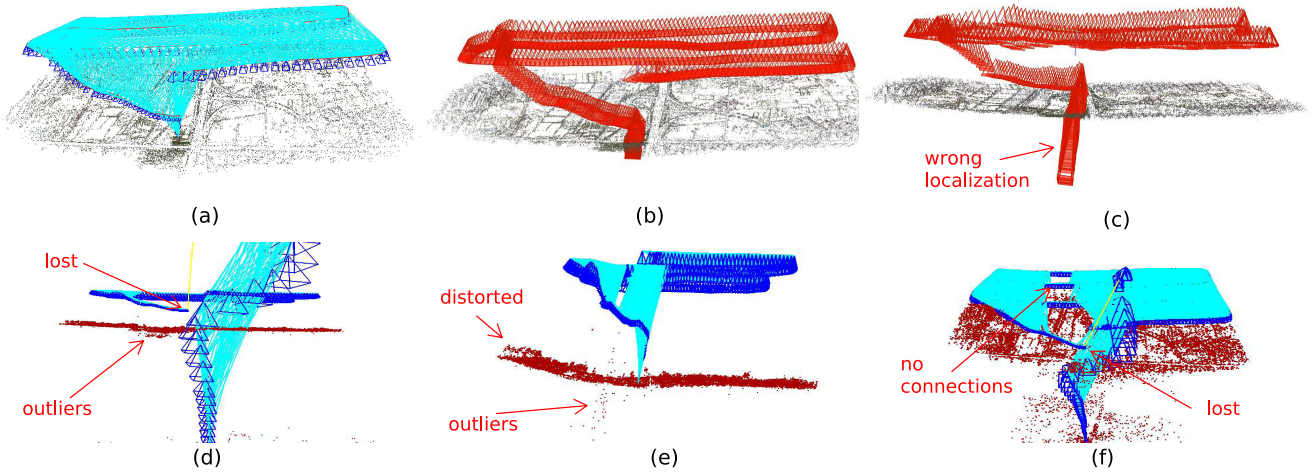


Fig. 6. Trajectory and sparse reconstruction comparison of sequence *phantom3-freeway* between RTSfM, COLMAP (incremental SfM) and OpenMVG (global pipeline). Although RTSfM is a real-time incremental reconstruction system, the localization, and mapping quality can be better than novel offline SfM systems. Self-calibration in SfM is not always robust, and RTSfM brings more good neighbor connections benefit from the novel matching algorithm and data association pipeline. The reconstruction of OpenMVG contains more outliers, in order to compare the matching module, we replace the features computation step of OpenMVG to the implementation of RTSfM (with 1000 keypoints), the missing connections of OpenMVG indicate RTSfM can achieve better matching result. (a) RTSfM. (b) COLMAP (fixK). (c) COLMAP. (d) OpenMVG (fixK). (e) OpenMVG. (f) OpenMVG (1000kps+fixK).

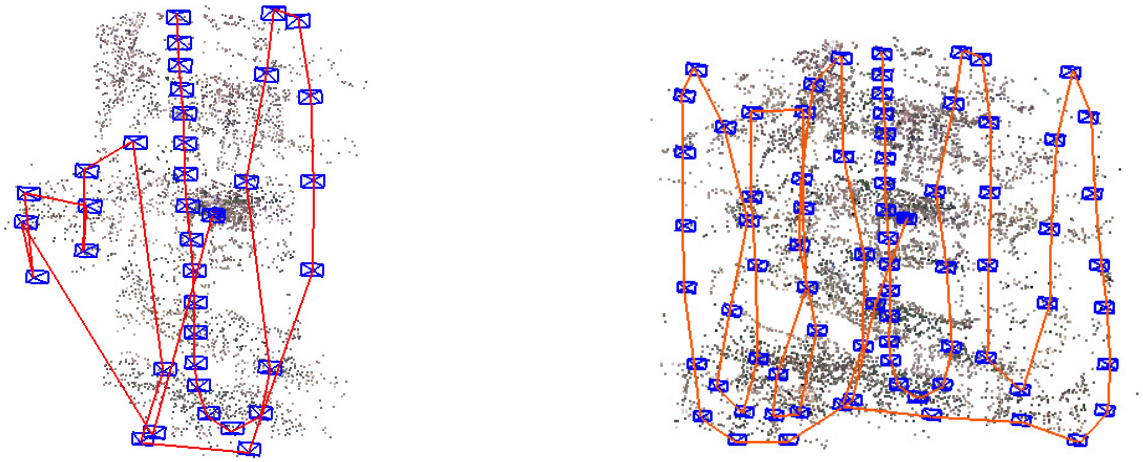


Fig. 7. Sparse reconstruction comparison with and without GPS when reduce the overlap by only use 1/5 keyframes of sequence *phantom3-village*. By jointly considering both GPS information and epiolar constraints, our tracking system is able to overcome overlap below 50% where PnP constraints are not available. (a) Without GPS. (b) With GPS.

IV. EXPERIMENTS

In this section, we perform the qualitative and quantitative evaluations of our approach. First, to evaluate the pose precision, we compare the camera trajectory of our approach with the well-known SLAM system ORBSLAM2 and open-source SfM projects COLMAP and OpenMVG on the DroneMap dataset, where high- and low-overlap image sequences are available. Second, to evaluate the precision of the online

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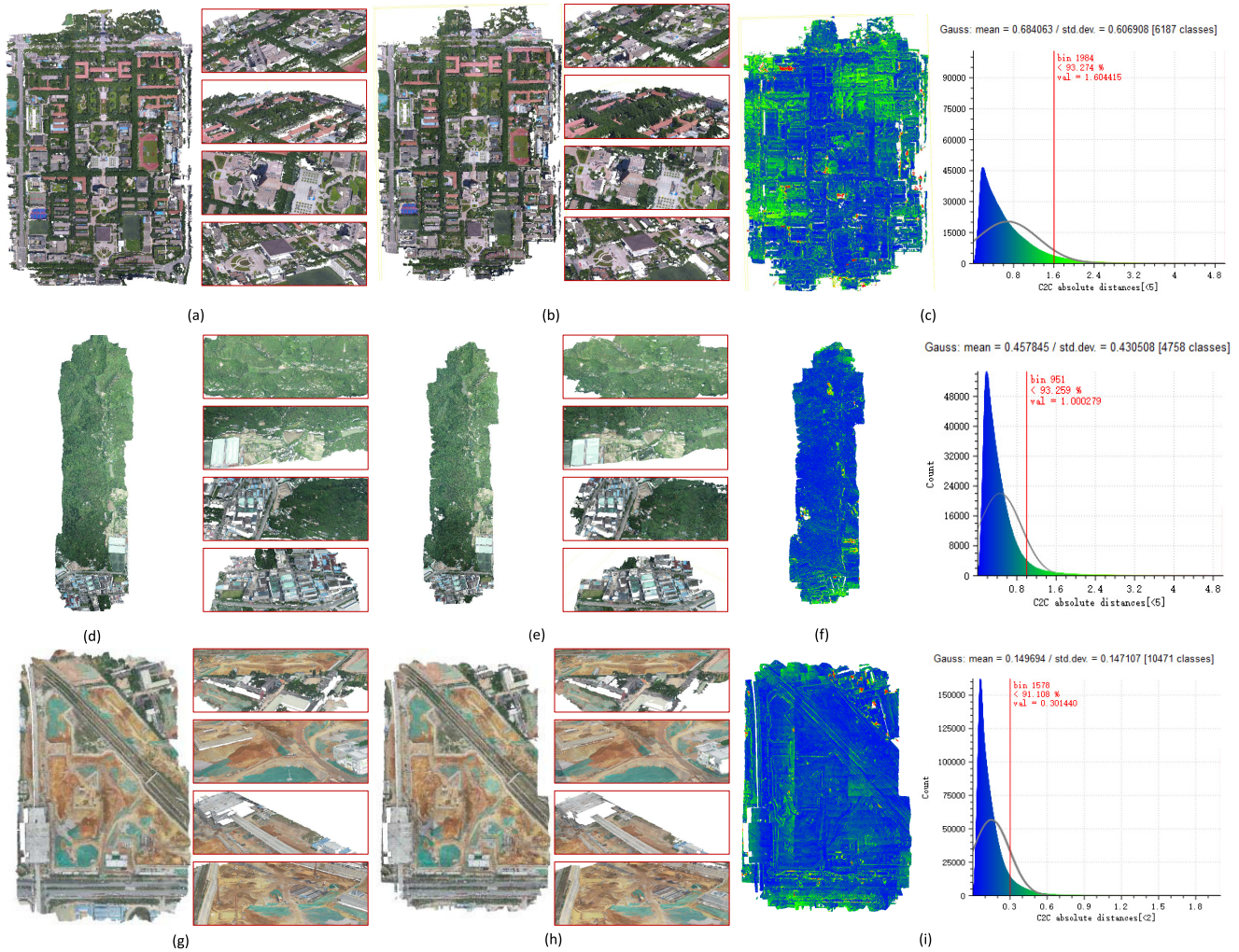


Fig. 8. Dense pointcloud comparison between Pix4DMapper (left) and RTSfM (middle), where iterative closest point (ICP) aligned residuals is illustrated (right). For Pix4DMapper we use the high-quality reconstruction, and RTSfM uses the online dense pointcloud result which is directly saved after all images tracked without global BA or extra stereo procedures. Sequence *mavic-campus* uses a low precision GPS and the Gauss mean difference is only 0.68 m even when there are lots of buildings. The average residual is 0.45 m and over 93.26% points are below 1 m, although the flight altitude of sequence *phantom4-mountain* is much higher and brings lower resolution [300 m with 8 cm ground sample distance (GSD)], since a higher precision RTK GPS is used. For sequence *p4r-mound*, we adopt a lower flight height and the average residual of ICP is below 0.15 m, where 91% points have less than 0.3 m error. (a) Pix4DMapper (mavic-campus). (b) Ours (mavic-campus). (c) ICP aligned residuals and histogram (mavic-campus). (d) Pix4DMapper (phantom4-moutain). (e) Ours (phantom4-moutain). (g) Pix4DMapper (p4r-mound). (h) Ours (p4r-mound). (i) ICP aligned residuals and histogram (p4r-mound).

dense pointcloud, we use the public DroneMap2[2] dataset, which is aimed by Pix4DMapper and treat its pointcloud as ground-truth. The pointcloud is aligned, and differences are illustrated with software CloudCompare. Third, we compare our ortho-mosaic and elevation image with Map2DFusion, TerrainFusion, and OpenREALM, while capturing a dataset with ground control points (GCPs) to evaluate the absolute precision. Finally, we compare the computational performance of our system with Pix4DMapper software to show our efficiency.

A. Pose Evaluation

The first output of our system is a sparse reconstruction that consists of map-points and key-frames, which are also the output of SLAM and SfM systems. To evaluate the quality of our reconstruction, we try to calculate the pose precision by align poses against the known GPS trajectory and compare with the state-of-the-art SLAM and SfM systems. ORB-SLAM2 [29] is a well-known SLAM system that claims high robustness and quality, whereas COLMAP [8] and OpenMVG [10] are two popular open-source SfM systems. Here COLMAP is a state-of-the-art incremental SfM system and OpenMVG provides both incremental and global reconstruction pipelines (global is used in this article).

The public aerial mapping dataset Northwestern Polytechnical University (NPU) DroneMap [1] with GPS trajectory is used to compare the above SLAM and SfM methods. Since GPS information will be used for pose evaluation, only the images with calibration information are used in the reconstructions for all methods. While RTSfM and ORB-SLAM2 use the given camera calibration, we evaluate both SfM results with or without intrinsic refinement, which is already joint optimized with GPS information. After the trajectories obtained, we use the evo tool (https://github.com/MichaelGrupp/evo) for trajectory evaluation. As illustrated in Fig. 5, most trajectories are all

[2]http://zhaoyong.adv-ci.com/npu-dronemap2-dataset

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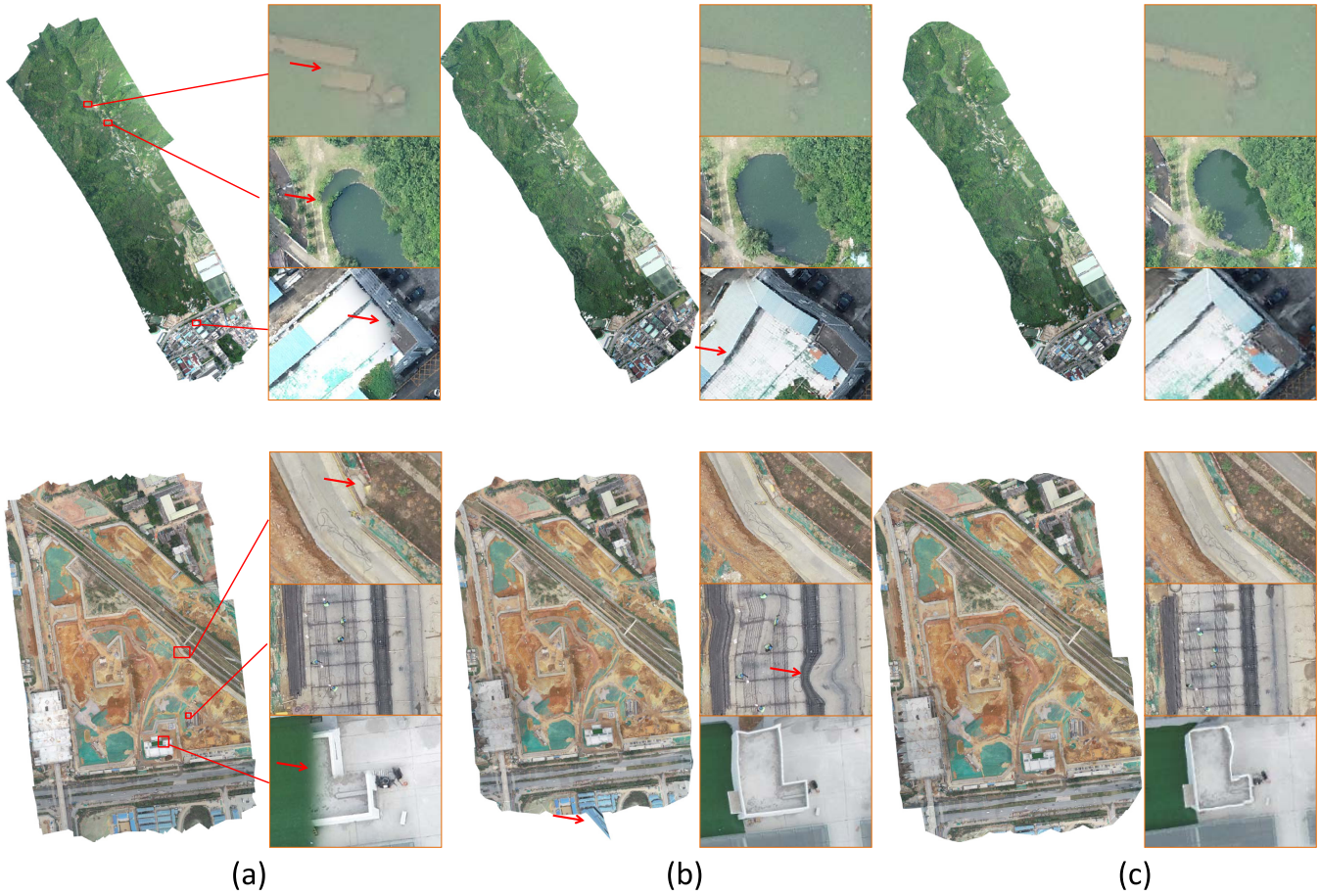


Fig. 9. Ortho-mosaic comparison of Map2DFusion [1], TerrainFusion [39] and RTSfM. Some differences are highlighted with red arrows. The results show that Map2DFusion contains mismatches because of planar assumption, and the mosaic is not a true-ortho because images are wrapped using homography projection. TerrainFusion is able to handle nonplanar scenes, such as mountains. However, the mosaic is highly twisted because the sparse pointcloud represent the 3-D structure details insufficiently, and although lot of filtering procedures are taken, outliers still existed. RTSfM can output the true ortho-mosaic based on the dense pointcloud and show the best result in areas with 3-D structures. (a) Map2DFusion. (b)TerrainFusion. (c) RTSfM (Ours).

aligned to the ground truth properly. The absolute trajectory error (ATE) statistic is listed in Table I. When we fix the camera intrinsic, the incremental SfM system COLMAP outputs the best results in most sequences because of its offline pipeline and final global BA, which is unrealistic for online methods. Our system obtains better trajectory precision compared with ORB-SLAM2, whereas the absolute position error is approximately $1/1000$, considering the flight area without the help of GPS. And we tried to remove our multihomography rematching strategy and use only BoW-based matching, the ablation experiment shows that our rematching strategy helps achieve better precision with the increase of matching recall.

What is more, our system obtains even better result in some sequences such as *phantom3-freeway* than SfM systems. Our data-association in mapping module finds more potential matches by projecting map-points, whereas SfM performs pairwise matching before the reconstruction. The detailed reconstruction results of RTSfM, COLMAP, and OpenMVG on this sequence are illustrated in Fig. 6, and the comparison shows that self-calibration in SfM is not always robust, which leads to wrong localizations and reconstruction distortions, we guess it is because most of our image sequences are keeping down-look and the camera intrinsics are hard to fix. Global pipeline of OpenMVG seems to output more outliers and the tracking fails at some point. To compare the matching module, we replace the features computation step of OpenMVG to the implementation of RTSfM (with 1000 keypoints), the missing connections of OpenMVG indicate RTSfM can achieve better matching result.

To further demonstrate the effectiveness of our large-baseline tracking system, we compare the sparse reconstructions with and without GPS and reduce the overlap by only using 1/5 keyframe images of sequence *phantom3-village* in Fig. 7. The result demonstrates that tracking fails when there is no GPS used, by jointly considering both GPS information and epipolar constraints, our tracking system is able to overcome overlap below 50% where PnP constraints are not available. The use of GPS not only prevents localization drift, provides reconstruction scale and georeference information, but also provides better robustness.

B. Dense Point-Cloud Evaluation

Traditional MVS dense pointcloud generation pipelines used in softwares, such as Pix4DMapper, are too slow and not suitable for our online framework where efficiency is required.

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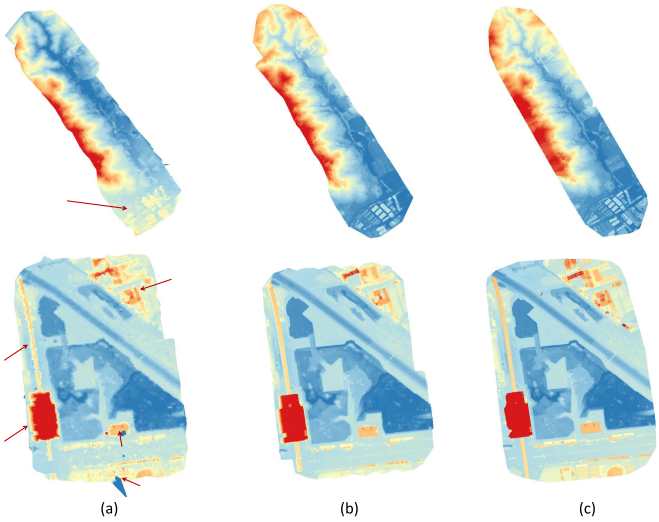


Fig. 10. DEM comparison of TerrainFusion [39], RTSfM and Pix4DMapper. The height is illustrated with pseudo color while red means high altitude and blue means low altitude. TerrainFusion is hard to describe the 3-D detail and the elevation image is not sharp enough especially for buildings and roads. RTSfM is able to provide high-quality elevation model which is comparable to that of Pix4DMapper. (a) TerrainFusion. (b) RTSfM (Ours). (c) Pix4DMapper.

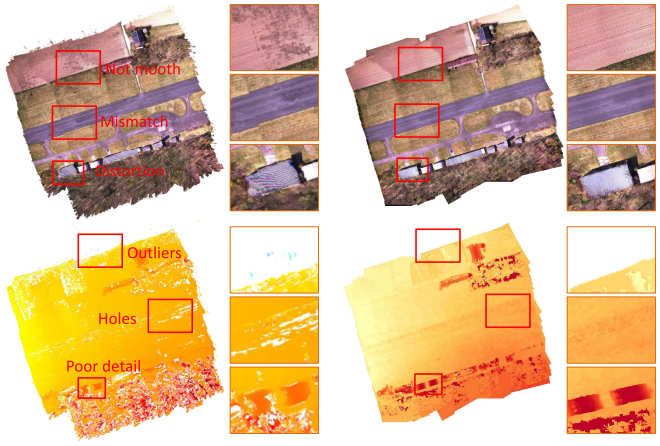


Fig. 11. Ortho-mosaic and elevation comparison between OpenREALM (left) and RTSfM (right). The comparison illustrates RTSfM not only provides smoother results with less outliers, mismatch, and distortions, but also fills all holes with inverse distance weighting (IDW) algorithm and reconstructs more details.

Our stereo and fusion module is able to output high-quality pointcloud results in real time. Although outperforming is challenging in terms of precision, several works are taken to improve the point cloud quality. To evaluate the quantitative precision of our dense reconstruction, we use the open-source cloud comparison software called CloudCompare[3] to illustrate the difference between our result and that of Pix4DMapper. We use the high-quality configuration setup for Pix4DMapper because we treat the output of it as ground truth. Two point-clouds are aligned with ICP algorithm and the point-cloud of Pix4DMapper is used as a reference. We use the default distance function in CloudCompare where no local modeling is performed. The comparison between our online pointcloud and the final fused result of Pix4DMapper is demonstrated in Fig. 8. Here, we directly saved the point cloud after all images are tracked, and no global BA or additional stereo procedures are taken because RTSfM is an online method. For sequence *p4r-mound*, over 100 million points are fused to the final pointcloud and the binary *.ply* file is over 1.6 GB. The experiment demonstrates that our method achieves high-quality dense pointcloud which is very close to the output of Pix4DMapper. We can also clearly see the 3-D structures, such as the buildings in pointcloud details. Given that SfM requires all images to be prepared before reconstructing, RTSfM generates the dense pointcloud incrementally, and able to visualize the online result during the flight. This real-time pointcloud is precise enough for different applications, and some measurements can be performed directly using the dense cloud because every point is georeferenced by fusing GPS information.

C. DSM Fusion Evaluation

RTSfM not only outputs real-time dense pointcloud, but also online DSM. Here, we further evaluate the quality of our digital ortho model (DOM) and digital elevation model (DEM). We first compare our ortho-mosaic result with two state-of-the-art online stitching systems, namely, Map2DFusion [1] and TerrainFusion [39]. The comparison result is illustrated in Fig. 9, highlighting the differences. The experiments demonstrate the following findings:

1) *Homography Projection Is Not Suitable for Nonplanar Scenes, Such as Mountains and Buildings:* Given that Map2DFusion fits the stitching plane with sparse map-points and images are stitched by homography projection, mismatches may occur in places with different heights.
2) *Sparse Map-Points Is Not Enough to Represent 3-D Structures:* Considering that TerrainFusion uses sparse pointcloud for DSM generation, describing the 3-D details correctly is difficult. This finding means that ortho-photo may be twisted and brings low-geometry precision and robustness.
3) *The Dense Reconstruction Helps Orthogonality and Quality of Ortho-Mosaic:* As demonstrated in Fig. 9, Map2DFusion and TerrainFusion fail to output correct ortho-mosaic for high buildings. Thus, the side of the building is still visible. Mismatches may also occur in such areas because the building is treated as a plane in Map2DFusion, and map-points are too sparse to represent the building edges in TerrainFusion. As for RTSfM, mismatches are hardly found, and a true ortho-mosaic is generated.

The dense reconstruction not only helps ortho-mosaic, but also improves elevation quality. Given that Map2DFusion can only output ortho-photo, we further compare our DEM with TerrainFusion and Pix4DMapper. The elevation model is exported to a *.tif* raster file and visualized with QGIS.[4] The pseudo color illustrated comparison of DEM is demonstrated in Fig. 10. The result shows that our DEM describes more details, the mountains and buildings can be clearly seen, while

[3]http://www.cloudcompare.org/

[4]https://github.com/qgis/QGIS

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TABLE II

TIME USAGE STATISTICS IN SECONDS FOR DIFFERENT SEQUENCES. OUR METHOD IS MUCH FASTER THAN PIX4DMAPPER AND MEETS THE DEMAND FOR REAL-TIME RECONSTRUCTION

Sequence	Location	Images	Resolution	Height (m)	Area (km^2)	GSD(m)	Pix4D (s)	Ours (s)
mavic-hongkong	Hong Kong	288	4000*3000	151	0.328	0.05	4723.0	**156.0**
mavic-fengniao	Xi'an, Shaanxi	216	4000*3000	60	0.073	0.02	3303.0	**134.6**
mavic-library	Xi'an, Shaanxi	205	4000*3000	111	0.738	0.03	3786.0	**144.6**
mavic-npu	Xi'an, Shaanxi	119	4000*3000	213	2.091	0.14	1834.0	**90.8**
mavic-xjtu	Xi'an, Shaanxi	166	4000*3000	213	1.181	0.08	2458.0	**136.5**
phantom3-olathe	Olathe, USA	160	4000*3000	122	0.257	0.05	2514.0	**146.9**
phantom4-mountain	Shenzhen, Guangdong	81	4864*3648	300	0.929	0.08	1752.0	**69.3**
p4r-mound	Shenzhen, Guangdong	365	4864*3648	100	0.355	0.03	11677.0	**436.7**
phantom3-npu	Xi'an, Shaanxi	13,983	1920*1080	254	1.598	0.30	-	**459.2**

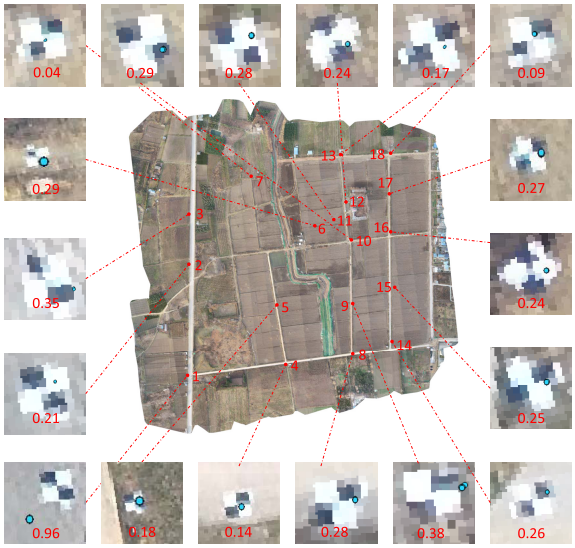


Fig. 12. Check GCP errors in *p4r-field* sequence where both GCPs and images contain high-quality RTK information. The white-black marker size is 600∗600 mm and blue dot means the GCP location. The absolute error means the distance between marker center and GCP position, which is measured in meters with red text. Since it is hard to fix the camera rotations in the early tracking stage, the first GCP contains a large residual and the later GCPs all have reasonable errors which are mostly below 0.3 m and enough for most applications.

TerrainFusion lacks visible details as no dense pointcloud generated. RTSfM can provide a high-quality elevation model that is comparable with that of Pix4DMapper. Furthermore, RTSfM does not need all images to be ready and does need hours of waiting, because all results are produced incrementally online. As OpenREALM [40] also implemented the elevation and ortho-mosaic fusing modules with densification, we further compared the result with it in Fig. 11. The comparison illustrates RTSfM not only provides smoother results with less outliers, mismatch and distortions, but also fills all holes with IDW algorithm and reconstructs more details. This is not only benefit from our localization system with smaller drift, but also the careful implementation of IDW-based elevation fusion algorithm and the adaptive weighted multiband ortho-mosaic fusing algorithm.

To further evaluate the quantitative precision of our reconstructed DOM, we use GCPs which are captured with a real-time kinematic (RTK) GPS for absolute error checking. Fig. 12 demonstrates the detail residuals of each GCP in sequence *p4r-field*, and the result shows that although RTSfM outputs a real-time ortho-mosaic without global optimization, the precision is good enough for most applications.

D. Efficiency Evaluation

Efficiency is our top priority because our approach aims to implement a real-time version of SfM. We compare the time usages of our system with those of Pix4DMapper to demonstrate the computational performance of our system. The time statistics for different sequences in seconds of RTSfM and Pix4DMapper are demonstrated in Table. II. All experiments are conducted on the same computer with Intel i7-6700 CPU, 16 GB RAM, and Nvidia GTX 1060 GPU. Pix4DMapper runs in Windows10 system while RTSfM runs in Mint 18 with Ubuntu 16.04 core. The result demonstrates that our system is over 20 times faster than Pix4DMapper in terms of tracking, mapping, dense reconstruction, and DSM fusion and generally processes one high-resolution image in less than a second. Generally, UAVs take one photo every two or more seconds. Thus, our system is fast enough for online high-resolution image processing. And our system is still able to handle video datasets in high frequency, sequence *phantom3-npu* demonstrates that our algorithm can process 1080 p video at over 30 Hz.

V. CONCLUSION

This article presents a novel approach to reconstruct sparse 3-D structures, dense point cloud, and DSM incrementally in real time with high efficiency and quality. High robustness feature matching and tracking pipeline are beneficial for small baseline video and high-resolution images that are supported to obtain high-precision camera poses. A stereo rectification-based method is used to generate image depth, and a high-quality dense point cloud is fused incrementally with further adjustment ability. Furthermore, an incremental DSM with true ortho-mosaic and elevation is fused and visualized with an online LOD terrain rendering and caching algorithm based on dense reconstruction. Considering that most traditional SfM methods need all data to be prepared and require hours of computation, the proposed method outputs comparative results with real-time speed and online visualization.

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It is worth noting that the precision of GPS is very important in our system since we tightly couple GPS in our tracking system. We generally prefer a Drone with RTK information so that the images can better associate with GPS information, and the synchronization of sensors is also important. A good calibration is also demanded like other existing multisensor fusion systems such as visual-inertial odometry (VIO).

We have already successfully deployed the algorithm in the DJI M300 drone with a Manifold 2C (i7-8500u), and only the live results are transmitted through the restful application programming interface (API). However, our implementation still need high computation cost, and in future works, more work should be conducted to port the algorithm to existing low-power ARM-based processors.

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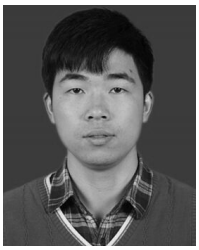
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